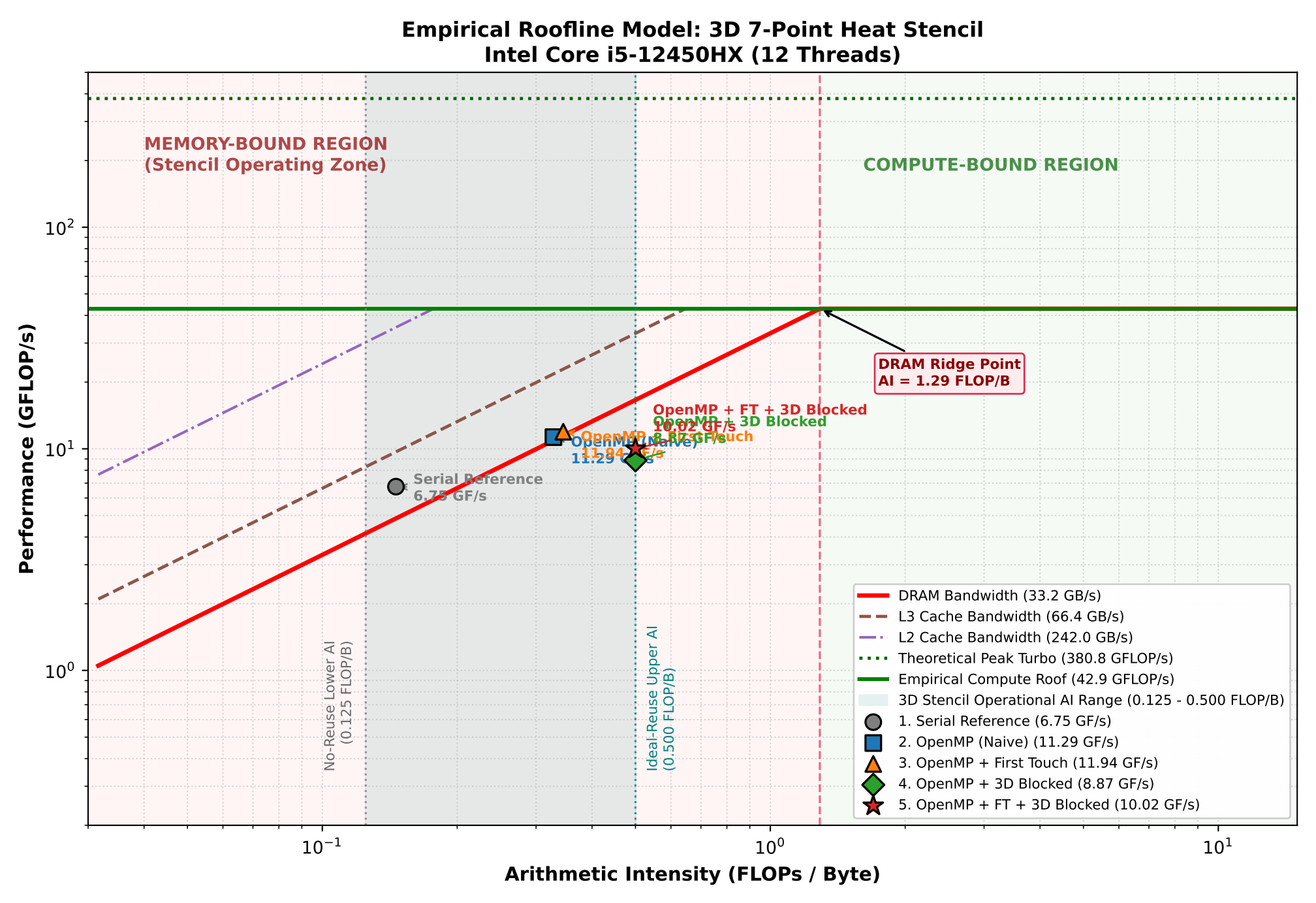
Empirical Roofline Model: 3D 7-Point Heat Stencil
Intel Core i5-12450HX (12 Threads)
MEMORY-BOUND REGION
(Stencil Operating Zone)
COMPUTE-BOUND REGION
DRAM Ridge Point
AI = 1.29 FLOP/B
OpenMP + FT + 3D Blocked
10.02 GF/s
OpenMP + 3D Blocked
OpenMP + First Touch
OpenMP (Naive)
11.29 GF/s
Serial Reference
6.75 GF/s
No-Reuse Lower AI
(0.125 FLOP/B)
Ideal-Reuse Upper AI
(0.500 FLOP/B)
Performance (GFLOP/s)
Arithmetic Intensity (FLOPs / Byte)
DRAM Bandwidth (33.2 GB/s)
L3 Cache Bandwidth (66.4 GB/s)
L2 Cache Bandwidth (242.0 GB/s)
Theoretical Peak Turbo (380.8 GFLOP/s)
Empirical Compute Roof (42.9 GFLOP/s)
3D Stencil Operational AI Range (0.125 - 0.500 FLOP/B)
1. Serial Reference (6.75 GF/s)
2. OpenMP (Naive) (11.29 GF/s)
3. OpenMP + First Touch (11.94 GF/s)
4. OpenMP + 3D Blocked (8.87 GF/s)
5. OpenMP + FT + 3D Blocked (10.02 GF/s)